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Fabrication of field effect transistors with ferroelectric materials

Abstract

A semiconductor structure includes gate spacers disposed over a semiconductor layer, a hafnium-containing dielectric layer, where a first portion of the hafnium-containing dielectric layer having a first thickness is disposed over the semiconductor layer and a second portion of the hafnium-containing dielectric layer having a second thickness is disposed along sidewalls of the gate spacers, and where the first thickness is greater than the second thickness, and a metal gate electrode disposed over the hafnium-containing dielectric layer and between the gate spacers.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a divisional of U.S. patent application Ser. No. 16/454,854, filed Jun. 27, 2019, the entire disclosure of which is incorporated herein by reference.

BACKGROUND

(1) Integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs, where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs.

(2) Such scaling down has also increased the complexity of processing and manufacturing ICs and, for these advances to be realized, similar developments in IC processing and manufacturing are needed. For example, high-k metal gates (HKMGs) have been fabricated with various advanced materials to improve speed and reliability of the ICs at reduced length scales. Although methods for forming HKMGs have been generally adequate, they have not been entirely satisfactory in all aspects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 is a flowchart of an example method for making a semiconductor device in accordance with some embodiments of the present disclosure.
- (3) FIG. 2 illustrates a three-dimensional perspective view of an example semiconductor device in accordance with some embodiments of the present disclosure.
- (4) FIGS. 3, 5, 6, 7, 8, 9A, 9B, 9C, 9D, 12, 14A, 14B, 14C, 14D, and 14E illustrate cross-sectional views of the semiconductor device of FIG. 2 taken along line AA' at various stages of the method of FIG. 1 in accordance with some embodiments of the present disclosure.
- (5) FIGS. 4 and 13 illustrate cross-sectional views of the semiconductor device of FIG. 2 taken along line BB' in accordance with some embodiments of the present disclosure.
- (6) FIGS. 10A and 10B illustrate embodiments of an example apparatus in accordance with some embodiments of the present disclosure.
- (7) FIGS. 10C and 10D illustrate embodiments of an example apparatus in accordance with some embodiments of the present disclosure.
- (8) FIG. 11 is a schematic representation of an example phase transformation process in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

- (9) The following disclosure provides many different embodiments, or examples, for implementing different features of the disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
- (10) In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. Moreover, the formation of a feature on, connected to, and/or coupled to another feature in the present disclosure that follows may include embodiments in which the features are formed in direct contact, and may also include embodiments in which additional features may be formed interposing the features, such that the features may not be in direct contact. In addition, spatially relative terms, for example, "lower," "upper," "horizontal," "vertical," "above," "over," "below," "beneath," "up," "down," "top," "bottom," etc. as well as derivatives thereof (e.g., "horizontally," "downwardly," "upwardly," etc.) are used for case of the present disclosure of one features relationship to another feature. The spatially relative terms are intended to cover different orientations of the device including the features. Still further, when a number or a range of numbers is described with "about," "approximate," and the like, the term is intended to encompass numbers that are within a reasonable range including the number described, such as within $\pm 10\%$ of the number described or other values as understood by person skilled in the art. For example, the term "about 5 nm" encompasses the dimension range from 4.5 nm to 5.5 nm.

(11) The present disclosure is generally related to semiconductor devices and the fabrication thereof, and more particularly to methods of fabricating high-k metal gates (HKMGs) in semiconductor devices, such as field-effect transistors (FETs). Throughout the present disclosure, “high-k” refers to a dielectric material having a dielectric constant k greater than that of silicon oxide, which is approximately 3.9.

(12) Embodiments such as those described herein provide methods of forming HKMGs having a metal gate electrode disposed over a high-k gate dielectric layer. In particular, the present disclosure provides methods of forming ferroelectric high-k gate dielectric layer in HKMGs. Generally, ferroelectric high-k dielectric materials have been employed to allow devices (e.g., FETs) to operate in a negative capacitance regime (e.g., in a negative-capacitance FET, or NCFET) for improved device performance. In one such example, ferroelectric high-k dielectric materials allow formation of FETs with reduced subthreshold swing (SS). SS generally describes the amount of voltage required to switch a device on and off, and thus influences the operating speed of the device. In many instances, other factors being constant, a reduction in SS generally increases a switching speed of an FET. SS may be controlled by the degree of ferroelectricity of a gate dielectric material included in a gate stack, with a higher ferroelectricity correlating to a lower SS. In addition, ferroelectric high-k dielectric materials may also enlarge the memory window for improved performance of nonvolatile memory devices. Notably, dielectric materials having similar compositions (e.g., hafnium-based high-k dielectric materials) may possess different degrees of ferroelectricity depending upon their specific crystalline phases (distinguished by different space groups, for example). Although methods of forming devices with enhanced ferroelectric properties have been generally adequate, they have not been satisfactory in all aspects.

(13) For example, crystalline HfO_2 in a high-k gate dielectric layer is generally obtained by thermally annealing amorphous or substantially amorphous HfO_2 to enable rearrangement of atoms within the amorphous phase and to form crystalline (e.g., orthorhombic) HfO_2 . However, as device sizes continue to decrease, the high-k gate dielectric layer with reduced thickness causes the energy barrier (e.g., activation energy for crystallization) required for phase transformation to increase such that the post-deposition thermal annealing alone energetically falls short in its ability to convert amorphous HfO_2 to orthorhombic crystalline HfO_2 . In one such example, it has been observed that when the thickness of the high-k gate dielectric layer reduces to about 5 nm and below, no amount or only a minimal amount of ferroelectric orthorhombic HfO_2 (i.e., a phase containing HfO_2 in the space group Pca2_1) may be formed after performing the thermal annealing process. One explanation for such phenomenon may be that as thickness reduces to length scales comparable to sizes of crystalline grains (e.g., within one order of magnitude in length scale), space available for the grains to form becomes inevitably limited, thereby presenting a greater energy barrier for crystallization process to occur.

(14) In addition, as device sizes continue to decrease, parasitic capacitance between the HKMG and gate spacers may inadvertently increase, compromising device performance. Furthermore, if thickness of the high-k gate dielectric layer disposed on the gate spacers is substantially the same as that disposed over the underlying substrate, the available window for tuning threshold voltage of the metal gate may shrink, and the overall gate resistance may also increase as a result.

(15) As will be discussed below, embodiments of the present disclosure are directed to methods of forming ferroelectric high-k gate dielectric layer in HKMGs at reduced length scales, addressing at least the concerns discussed above. In particular, methods of promoting transformation of amorphous HfO_2 to crystalline HfO_2 , and more specifically, to crystalline HfO_2 in a ferroelectric orthorhombic phase are provided herein. Additionally, the present disclosure provides methods of enhancing the overall performance of HKMGs in applications such as NCFETs.

(16) FIG. 1 illustrates a flow chart of a method **100** for forming a semiconductor device (hereafter referred to as “device”) **200** in accordance with some embodiments of the present disclosure. The method **100** is merely an example and is not intended to limit the present disclosure beyond what is

explicitly recited in the claims. Additional operations can be performed before, during, and after the method **100**, and some operations described can be replaced, eliminated, or moved around for additional embodiments of the method. The method **100** is described below in conjunction with FIGS. **2-9D** and **12-14E** which illustrate various three-dimensional and cross-sectional views of the device **200** during intermediate steps of the method **100**. In particular, FIG. **2** illustrates a three-dimensional view of the device **200**, while FIGS. **3, 5-9D, 12,** and **14A-14E** illustrate cross-sectional views of the device **200** taken along line AA' as shown in FIG. **2**, and FIGS. **4** and **13** illustrate cross-sectional views of the device **200** taken along line BB' as shown in FIG. **2**.

(17) The device **200** may be an intermediate device fabricated during processing of an integrated circuit (IC), or a portion thereof, that may comprise static random-access memory (SRAM) and/or other logic circuits, passive components such as resistors, capacitors, and inductors, and active components such as p-type FETs (PFETs), n-type FETs (NFETs), fin-like FETs (FinFETs), metal-oxide semiconductor field effect transistors (MOSFET), complementary metal-oxide semiconductor (CMOS) transistors, bipolar transistors, high voltage transistors, high frequency transistors, and/or other memory cells. The present disclosure is not limited to any particular number of devices or device regions, or to any particular device configurations. For example, though the device **200** as illustrated is a three-dimensional FinFET device, the present disclosure may also provide embodiments for fabricating planar FET devices, nanosheets, nanowires, etc.

(18) Referring to FIGS. **1-4**, the method **100** at operation **102** provides the device **200** that includes one or more fins **204** protruding from a substrate **202** and separated by isolation structures **208**, a dummy gate stack **206** disposed over the fins **204**, gate spacers **212** disposed on sidewalls of the dummy gate stack **206**, epitaxial source/drain (S/D) features **214** disposed over the fins **204**, and an interlayer dielectric (ILD) layer **218** disposed over the isolation structures **208** and the epitaxial S/D features **214**. Though not depicted herein, the device **200** may include additional material layers suitable for various design requirements.

(19) The substrate **202** may comprise an elementary (single element) semiconductor, such as silicon, germanium, and/or other suitable materials; a compound semiconductor, such as silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, indium antimonide, and/or other suitable materials; an alloy semiconductor, such as SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, and/or other suitable materials. The substrate **202** may be a single-layer material having a uniform composition. Alternatively, the substrate **202** may include multiple material layers having similar or different compositions suitable for IC device manufacturing. In one example, the substrate **202** may be a silicon-on-insulator (SOI) substrate having a silicon layer formed on a silicon oxide layer. In another example, the substrate **202** may include a conductive layer, a semiconductor layer, a dielectric layer, other layers, or combinations thereof.

(20) In some embodiments where the substrate **202** includes FETs, various doped regions, such as source/drain regions, are disposed in or on the substrate **202**. The doped regions may be doped with n-type dopants, such as phosphorus or arsenic, and/or p-type dopants, such as boron, depending on design requirements. The doped regions may be formed directly on the substrate **202**, in a p-well structure, in an n-well structure, in a dual-well structure, or using a raised structure. Doped regions may be formed by implantation of dopant atoms, in-situ doped epitaxial growth, and/or other suitable techniques.

(21) Each fin **204** may be suitable for providing an n-type FinFET or a p-type FinFET. The fins **204** may be fabricated using suitable processes including photolithography and etch processes. The photolithography process may include forming a photoresist layer (resist) overlying the substrate **202**, exposing the resist to a pattern, performing post-exposure bake processes, and developing the resist to form a masking element (not shown) including the resist. The masking element is then used for etching recesses into the substrate **202**, leaving the fins **204** on the substrate **202**. The etching process may include dry etching, wet etching, reactive ion etching (RIE), and/or other

suitable processes.

(22) Numerous other embodiments of methods for forming the fins **204** may be suitable. For example, the fins **204** may be patterned using double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the fins.

(23) The isolation structures **208** may include silicon oxide, silicon nitride, silicon oxynitride, fluoride-doped silicate glass (FSG), a low-k dielectric material, and/or other suitable materials. The isolation structures **208** may include shallow trench isolation (STI) features. In one embodiment, the isolation structures **208** are formed by etching trenches in the substrate **202** during the formation of the fins **204**. The trenches may then be filled with an isolating material described above by a deposition process, followed by a chemical mechanical planarization (CMP) process. Other isolation structure such as field oxide, local oxidation of silicon (LOCOS), and/or other suitable structures may also be implemented as the isolation structures **208**. Alternatively, the isolation structures **208** may include a multi-layer structure, for example, having one or more thermal oxide liner layers. The isolation structures **208** may be deposited by any suitable method, such as chemical vapor deposition (CVD), flowable CVD (FCVD), spin-on-glass (SOG), other suitable methods, or combinations thereof.

(24) In some embodiments, the dummy gate stack **206** is provided as a placeholder for an HKMG and may include a dummy gate electrode comprising polysilicon. The dummy gate stack **206** may additionally include other material layers disposed between the dummy gate electrode and the fins **204**. For example, the dummy gate stack may include an interfacial layer (not depicted) and/or a dummy gate dielectric layer (not depicted). As will be discussed in detail below, at least portions of the dummy gate stack **206** are replaced with the HKMG during a gate replacement process after other components (e.g., the epitaxial S/D features **214**) of the device **200** are fabricated. Various material layers of the dummy gate stack **206** may be formed by any suitable process, such as CVD, physical vapor deposition (PVD), atomic layer deposition (ALD), chemical oxidation, other suitable processes, or combinations thereof.

(25) The device **200** further includes gate spacers **212** disposed on sidewalls of the dummy gate stack **206**. The gate spacers **212** may include a dielectric material, such as silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, other suitable dielectric materials, or combinations thereof. The gate spacers **212** may be a single layered structure or a multi-layered structure. The gate spacers **212** may be formed by first depositing a blanket of spacer material over the device **200**, and then performing an anisotropic etching process to remove portions of the spacer material to form the gate spacers **212** on sidewalls of the dummy gate stack **206**. In the present disclosure, the gate spacers **212** remain as portions of the device **200** during the subsequent gate replacement process.

(26) Collectively referring to FIGS. 2 and 3, the device **200** includes epitaxial S/D features **214** disposed over the fins **204** and adjacent to the dummy gate stack **206**. The epitaxial S/D features **214** may be formed by any suitable techniques, such as etching processes followed by one or more epitaxy processes. In one example, one or more etching processes are performed to remove portions of the fins **204** to form recesses (not depicted) therein, respectively. A cleaning process may be performed to clean the recesses with a hydrofluoric acid (HF) solution or other suitable solution. Subsequently, one or more epitaxial growth processes are performed to grow epitaxial features in the recesses. Each of the epitaxial S/D features **214** may be suitable for a p-type FinFET device (e.g., a p-type epitaxial material) or alternatively, an n-type FinFET device (e.g., an n-type

epitaxial material). The p-type epitaxial material may include one or more epitaxial layers of silicon germanium (epi SiGe), where the silicon germanium is doped with a p-type dopant such as boron, germanium, indium, and/or other p-type dopants. The n-type epitaxial material may include one or more epitaxial layers of silicon (epi Si) or silicon carbon (epi SiC), where the silicon or silicon carbon is doped with an n-type dopant such as arsenic, phosphorus, and/or other n-type dopant.

(27) Additionally, the device **200** includes the ILD layer **218**, and optionally a contact etch-stop layer (CESL; not depicted), disposed over the substrate **202**. In some embodiments, the ILD layer **218** includes a dielectric material, such as tetraethylorthosilicate (TEOS), un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), other suitable dielectric materials, or combinations thereof. The ILD layer **218** may include a multi-layer structure having multiple dielectric materials and may be formed by a deposition process such as, for example, CVD, FCVD, SOG, other suitable methods, or combinations thereof. In some embodiments, forming the ILD layer **218** further includes performing a CMP process to planarize a top surface of the device **200**, such that a top surface of the dummy gate stack **206** is exposed. The CESL may comprise silicon nitride, silicon oxynitride, silicon nitride with oxygen or carbon elements, other suitable materials, or combinations thereof, and may be formed by CVD, PVD, ALD, other suitable methods, or combinations thereof.

(28) Now referring to FIGS. **1** and **5**, the method **100** at operation **104** removes the dummy gate stack **206** to form a gate trench **220**. In the depicted embodiment, the gate trench **220** exposes portions of the fins **204** and the isolation structures **208**. The method **100** at operation **104** may include one or more etching processes that are selective to the materials included in the dummy gate stack **206** (e.g., polysilicon). The etching processes may include dry etching, wet etching, RIE, or other suitable etching methods, or combinations thereof.

(29) Referring to FIGS. **1**, **6**, and **7**, the method **100** at operation **106** forms a sacrificial layer **230** over the gate spacers **212**. The method **100** forms the sacrificial layer **230** in a selective manner over the gate spacers **212** such that the sacrificial layer **230** physically and/or chemically bonds with or grows on the gate spacers **212** (e.g., silicon nitride, silicon oxide, etc.) but not with or on the fin **204** or the epitaxial S/D features **214**. In some embodiments, the sacrificial layer **230** is formed to a thickness t ranging from about 0.5 nm to about 1 nm. In some embodiments, as depicted in FIG. **6**, the method **100** first implements a cleaning process **310** to prepare surfaces of the gate spacers **212** for the formation of the sacrificial layer **230**. In some embodiments, the cleaning process **310** may be implemented using a wet agent such as hydrofluoric acid (HF). In the present disclosure, the cleaning process **310** is configured to functionalize the surfaces of the fin **204** with hydrogen-terminated (—H) groups and the surfaces of the gate spacers **212** with hydroxyl-terminated (—OH) groups. As such, chemical selectivity is established between the fin **204** and the gate spacers **212** to accommodate the selective deposition of the sacrificial layer **230**.

(30) In one example embodiment, the sacrificial layer **230** includes a polymer material in the form of a self-assembled monolayer (SAM) configured to provide various functional groups at surfaces of the gate spacers **212**. In some examples, the SAM may include a sulfur-containing functional group (e.g., thiols), a silicon-containing group (e.g., silanes), an oxygen-containing group (e.g., hydroxyls), a methyl-containing group, or combinations thereof. The SAM may be formed by any suitable deposition process such as, for example, spread coating, chemisorption, PVD, electrodeposition, micro-contact printing, other suitable processes, or combinations thereof. In another example embodiment, the sacrificial layer **230** includes a layer of pristine graphene having a few, if any, structural defects. The layer of graphene may be formed by any suitable deposition process such as CVD. As provided herein, portions of the sacrificial layer **230** are configured to be removed at a later processing step.

(31) In some embodiments, the method **100** may subsequently form various material layers in the

gate trench **220**. For example, referring to FIG. **8**, the method **100** may form an interfacial layer **222** in the gate trench **220** to any suitable thickness. The interfacial layer **222** may be formed by any suitable method such as, for example, thermal oxidation, chemical oxidation, CVD, ALD, PVD, other suitable methods, or combinations thereof. Notably, due to the presence of the sacrificial layer **230** over the gate spacers **212**, the interfacial layer **222** selectively bonds with or grows on the fin **204** but not on the sacrificial layer **230** (though the interfacial layer **222** is physically disposed adjacent to the sacrificial layer **230**). In some embodiments, the interfacial layer **222** is omitted from the device **200**.

(32) Thereafter, the method **100** may proceed in one of two paths, Path A and Path B. The method **100** including Path A is discussed with reference to FIGS. **9A** to **9D**, while the method **100** including Path B is discussed with reference to FIGS. **14A** to **14E**. For purposes of clarity, numeric designations for components of the device **200** will remain the same during the discussion of Path A and Path B.

(33) Referring to FIGS. **1** and **9A**, the method **100** at operation **108A** forms a Hf-containing layer **232A** in the gate trench **220**. Notably, as depicted herein, operation **108A** simultaneously deposits the Hf-containing layer **232A** over the fin **204** while applying an electric field **320** over the device **200**. The Hf-containing layer **232A** is selectively formed over the fin **204** (or any material layers formed thereover, such as the interfacial layer **222**), as the sacrificial layer **230** includes surface termination groups not compatible for bonding with the Hf-containing layer **232A**. In other words, the sacrificial layer **230** facilitates or guides the selective deposition of the Hf-containing layer **232A** over portions of the fin **204** (or any material layers formed thereover) rather than over surfaces of the gate spacers **212**.

(34) The Hf-containing layer **232A** includes a high-k dielectric material comprising hafnium (Hf) and oxygen (O) in the form of, for example, hafnium oxide (e.g., HfO₂). Therefore, in the present disclosure, the Hf-containing layer **232A** may alternatively be referred to as the Hf-containing gate dielectric layer **232A**. As discussed above, “high-k” refers to a dielectric material having a dielectric constant greater than that of silicon oxide, which is about 3.9. In some examples, an amount (by wt %) of HfO₂ included in the Hf-containing layer **232A** ranges from about 30% to about 100%. In further embodiments, the Hf-containing layer **232A** is doped with one or more dopant including, but not limited to, zirconium (Zr), aluminum (Al), lanthanum (La), titanium (Ti), tantalum (Ta), silicon (Si), yttrium (Y), scandium (Sc), other suitable elements, or combinations thereof.

(35) In the present embodiments, the Hf-containing layer **232A** primarily includes amorphous HfO₂ rather than crystalline HfO₂, rendering the Hf-containing layer **232A** having little to no ferroelectricity. Besides amorphous HfO₂, the Hf-containing layer **232A** may also include a minute amount of microcrystals of HfO₂, which similarly does not exhibit ferroelectric behavior. In one example, the Hf-containing layer **232A** may include more than about 90% (by weight) of amorphous HfO₂, and about 5% of the remainder may be crystalline HfO₂ in a monoclinic phase.

(36) The Hf-containing layer **232A** may be formed by any suitable deposition process including ALD, CVD, other processes, or combinations thereof. In some examples, the deposition of the Hf-containing layer **232A** may be performed at a temperature ranging from about 150 degrees Celsius to about 500 degrees Celsius depending upon the specific layer thickness desired and/or if any dopant(s) is included. In some examples, the Hf-containing layer **232A** may be formed at a gas pressure ranging from about 0.5 Torr to about 3 Torr depending upon the specific layer thickness desired and/or if any dopant(s) is included.

(37) The Hf-containing layer **232A** may optionally include other high-k dielectric material, such as alumina (Al₂O₃), zirconium oxide (ZrO₂), lanthanum oxide (La₂O₃), titanium oxide (TiO₂), yttrium oxide (Y₂O₃), other suitable high-k materials, or combinations thereof. In some embodiments, the Hf-containing layer **232A** may include a

perovskite material, such as lead zirconate titanate ($\text{PbZr}_{0.5}\text{Ti}_{0.5}\text{O}_3$, or PZT, where $0 \leq x \leq 1$), barium titanate (BaTiO_3), barium strontium titanate (BaSrTiO_x , or BST), strontium titanate (SrTiO_3), other suitable perovskite materials, or combinations thereof that are configured to undergo phase transition to acquire ferroelectric behavior.

(38) Still referring to FIG. 9A, the in-situ application of the electric field 320 may be performed in the same chamber as the deposition process implemented at operation 108A. Referring to FIGS. 10A and 10B, a deposition apparatus 400 is provided as an example embodiment in which the deposition of the Hf-containing layer 232A and/or the application of the electric field 320 is implemented.

(39) The deposition apparatus 400 includes a chamber 402 configured with two electrodes: a top electrode 404 and a bottom electrode 406. In some embodiments, the bottom electrode 406 includes an electrostatic chuck (or E-chuck) upon which the device 200 is disposed. Alternatively or additionally, the bottom electrode 406 may include a susceptor, which is configured to heat the device 200 disposed thereon to an elevated temperature using, for example, infrared radiation. In some embodiments, referring to FIG. 10A, the top electrode 404 and the bottom electrode 406 are configured to be substantially parallel. However, the present disclosure is not limited to this configuration and may also be applicable in instances where the top electrode 404 and/or the bottom electrode 406 is configured with a curvature (e.g., FIG. 10B), such as in a concave, convex, spiral, or other suitable configurations. In some examples, spacing between the top electrode 404 and the bottom electrode 406 ranges from about 5 cm to about 50 cm; though other spacings may also be applicable in order to accommodate different configurations and sizes of the deposition apparatus 400. The top electrode 404 and the bottom electrode 406 may include any suitable metal or ceramic materials operable to be heated to a desired temperature (e.g., from about 150 degrees Celsius to about 500 degrees Celsius as discussed above). The deposition apparatus 400 may further include a gas inlet 416 and a gas outlet (not depicted) for providing one or more gas, inert or otherwise, during deposition, temperature monitoring device(s), and/or other suitable components. In some examples, gases such as nitrogen (N_2), argon (Ar), ammonia (NH_3), oxygen (O_2), other suitable gases, or combinations thereof may be provided to the chamber 402 via the gas inlet 416.

(40) Referring to both FIGS. 10A and 10B, the top electrode 404 and the bottom electrode 406 are coupled to a power source 408, which may be disposed inside the chamber 402 as depicted herein or, alternatively, disposed outside the chamber 402. When the power source 408 is switched on, the top electrode 404 and the bottom electrode 406 may be oppositely charged such that an electric field 412 (e.g., the electric field 320) is established therebetween. For embodiments in which the top electrode 404 and the bottom electrode 406 are substantially parallel to each other (FIG. 10A), a direction of the electric field 412 may be substantially unidirectional, such as substantially perpendicular to both the top electrode 404 and the bottom electrode 406. Alternatively, referring to FIG. 10B, the direction of the electric field 412 may be multidirectional, i.e., the electric field 412 does not align along a single direction. In the present disclosure, the electric field 320 is substantially unidirectional and may be substantially perpendicular to a top surface of the device 200 disposed over the bottom electrode 406. In some embodiments, the electric field 412 is a direct-current (DC) electric field. In alternative embodiments, the electric field 412 is an alternating-current (AC) electric field. Furthermore, the polarity of the electrodes 404 and 406 may be reversed for applications of the present disclosure. In some examples, the voltage supplied by the power source 408 is from about 2 kV to about 10 kV; though other voltage may also be applicable in the present disclosure. During the deposition process 250, a source material 414 is discharged into the chamber 402 through a dispensing unit 410 and deposited on the device 200 (e.g., in the gate trench 220). In the depicted embodiment, the source material 414 includes one or more precursor gas suitable for forming HfO_2 . However, the present disclosure is not limiting with respect to the kinds of precursor gas that may be used herein. For example, the precursor gas

may include an Hf-containing metalorganic gas, an Hf-containing halide gas, other suitable gases, or combinations thereof.

(41) In the present embodiments, the in-situ application of the electric field **320** and the deposition of the Hf-containing layer **232A** provides energy to mobilize oxygen atoms relative to hafnium atoms toward lattice positions consistent with the ferroelectric orthorhombic crystal structure. In some instances, energy provided by the electric field **320** may be utilized to control the amount of oxygen vacancies present in the crystal structure. In some embodiments, a suitable amount of oxygen vacancies helps facilitate the phase transition to crystalline phases with high symmetry, such as orthorhombic phase, during a subsequent annealing process and induce metastable polar phases to form at the boundary of such phases. However, an excessive amount of oxygen vacancies may instead favor less symmetric crystalline phases that do not exhibit ferroelectric properties. As a result, though not required, it may offer advantage for the overall yield of the ferroelectric orthorhombic phase to apply an electric field during the deposition of the Hf-containing layer **232A**, resulting in a Hf-containing layer with microstructures different (e.g., having less oxygen vacancies) from those of an Hf-containing layer not deposited with in-situ application of an electric field. Though not required by the present disclosure, elements such as nitrogen (N) (e.g., in the form of an inert gas provided during operation **108A**) may passivate the oxygen vacancies, further promoting the phase transformation as discussed herein.

(42) It is noted, however, that the energy provided by the electric field **320** during the deposition process at the operation **108A** is generally not sufficient to complete the transformation of amorphous HfO₂ to ferroelectric orthorhombic crystalline HfO₂, and a subsequent annealing process may be needed to facilitate such transformation. Additionally, as will be discussed with respect to Path B, the present disclosure does not require the in-situ application of an electric field during the deposition of the Hf-containing layer **232A**; instead, an electric field may be applied alternatively or additionally at a later stage during the method **100** to achieve enhanced transformation of amorphous HfO₂ to ferroelectric orthorhombic HfO₂.

(43) Thereafter, referring back to FIGS. **1** and **9B**, the method **100** at operation **110** removes the sacrificial layer **230** to form a trench **240**. In the depicted embodiment, the trench **240**, having a width substantially equivalent to the thickness t of the sacrificial layer **230**, is disposed between the Hf-containing layer **232A** and a sidewall of the gate spacers **212**. In some embodiments, the sacrificial layer **230** is removed by a wet etching process **330**. In some embodiments, implementing the wet etching process **330** includes applying an etchant that includes solvent such as sulfuric acid, hydrogen peroxide, other suitable solvents, or combinations thereof.

(44) Referring to FIGS. **1** and **9C**, the method **100** at operation **112A** forms an Hf-containing layer **232B** over the device **200**. Specifically, the method **100** forms a first portion of the Hf-containing layer **232B** over the Hf-containing layer **232A** and a second portion of the Hf-containing layer **232B** in the trench **240**. In the present embodiments, the method **100** forms the Hf-containing layer **232B** by implementing an ALD process. In the depicted embodiment, the Hf-containing layer **232A** and the Hf-containing layer **232B** include substantially the same material (e.g., amorphous HfO₂) and are therefore together referred to as Hf-containing layer **233** having a thickness T that is greater than the thickness t . In the depicted embodiment, the thickness T ranges from about 1 nm to about 3 nm. Notably, if T is less than about 1 nm, current leakage of the HKMG **210** may be too high, adversely affecting the performance of the device **200**. On the other hand, if T is greater than about 3 nm, dimensions of subsequently formed layers over the Hf-containing layer **233** may be negatively affected, e.g., less available space for forming a bulk conductive layer of the HKMG **210**. In some examples, a ratio of T to t may be at least about 3. Although the present disclosure is not limited to such ratio, it is generally desirable to minimize the thickness t (i.e., maximizing the ratio of T to t) to an extent allowable by the capacity of the processing tool(s) because a thinner Hf-containing layer **232B** may lead to greater reduction in parasitic capacitance of the device **200**. In some instances, if the ratio of T to t is less than about 3, the extent of reduction in parasitic

capacitance may be negligible. Due to the removal of the sacrificial layer **230**, the Hf-containing layer **232B** is formed over both the Hf-containing layer **232A** and sidewalls of the gate spacers **212** while filling up the trench **240**. As a result, the Hf-containing layers **232A** and **232B** include a high-k dielectric material, such as HfO₂, with controlled oxygen vacancies suitable for facilitating the transformation of amorphous HfO₂ to crystalline ferroelectric orthorhombic HfO₂.

(45) In some embodiments, the Hf-containing layer **232B** is formed by simultaneously depositing an Hf-containing material (e.g., HfO₂) and applying an electric field **340**. The process of forming the Hf-containing layer **232B** may be similar to the process of forming the Hf-containing layer **232A** at operation **108A**. For example, referring back to FIGS. **10A** and **10B**, the Hf-containing layer **232B** may be formed in an apparatus similar to the deposition apparatus **400**, in which a source material for HfO₂ is discharged into the chamber **402** through a dispensing unit **410** toward the substrate (e.g., the device **200**). The application of the electric field **340** may be implemented by switching on the power source **408** to charge the top electrode **404** and the bottom electrode **406** with opposite charges, thereby establishing an electric field therebetween. The direction of the electric field **340** may be unidirectional or multidirectional with respect to the device **200**. In some embodiments, the direction and strength (i.e., the magnitude) of the electric field **340** is substantially the same as those of the electric field **320** as discussed above.

(46) Notably, still referring to FIG. **9C**, the relative dimensions (e.g., a ratio) of the thickness *t* and a thickness *H*, which is a thickness of the Hf-containing layer **233** and the interfacial layer **222** combined, may be used to determine a specific means of deposition process that may be implemented at operation **112A**. For example, if the ratio of the thickness *H* to the thickness *t* is less than about 4, then a deposition process such as ALD may be used to form the Hf-containing layer **232B**. For ratios that are greater than about 4, both ALD and CVD may be implemented to form the Hf-containing layer **232B**. Of course, regardless of the specific means of deposition, the deposition apparatus **400** depicted in FIGS. **10A** and **10B** is generally applicable to the present embodiments to ensure simultaneous deposition of an Hf-containing material and application of an electric field.

(47) Thereafter, referring to FIGS. **1** and **9D**, the method **100** at operation **114** performs an annealing process with an in-situ application of an electric field **350** to the device **200**. As will be discussed below, the annealing process implemented at operation **114** alters the structure of the HfO₂ included in the Hf-containing layers **232A** and **232B** to form a Hf-containing layer **234** disposed over the device **200**. In the present disclosure, the Hf-containing layer **234** includes a first portion **234A** (or Hf-containing layer **234A**) that corresponds to the Hf-containing layer **233** disposed over the fin **204** and a second portion **234B** (or Hf-containing layer **234B**) that corresponds to the second portion of the Hf-containing layer **232B** disposed over the gate spacers **212** (see FIG. **9C**).

(48) In some embodiments, the method **100** at operation **114** performs the annealing process in an annealing apparatus **500** as depicted in FIGS. **10C** and **10D**. The annealing apparatus **500** includes a chamber **502** configured with two electrodes: a top electrode **504** and a bottom electrode **506**, which may be similar in configuration and/or in material composition as the top electrode **404** and the bottom electrode **406** discussed above with respect to the deposition apparatus **400**. For example, the top electrode **504** and the bottom electrode **506** may be configured to be substantially parallel as depicted in FIG. **10C** or, alternatively, the top electrode **504** and/or the bottom electrode **506** may be configured with a curvature as depicted in FIG. **10D**. The annealing apparatus **500** may additionally or alternatively include a heating source **510** disposed below and in contact with the bottom electrode **506**. The heating source **510** may include a susceptor as discussed above configured to heat the device **200** disposed thereover to an elevated temperature during the annealing process. In some embodiments, the bottom electrode **506** and the heating source **510** may be a single, unified component. Alternatively, the bottom electrode **506** and the heating source **510** may be configured to be separate components. The annealing apparatus **500** may include additional

components, such as a gas inlet **516**, which may be similar to the gas inlet **416** depicted in FIGS. **10A** and **10B**, and a gas outlet (not depicted), other suitable components, or combinations thereof. Gases such as nitrogen (N.sub.2), argon (Ar), ammonia (NH.sub.3), other suitable gases, or combinations thereof may be applied to the device **200** during the annealing process.

(49) Similar to the descriptions of FIGS. **10A** and **10B**, the top electrode **504** and the bottom electrode **506** are coupled to a power source **508**. When the power source **508** is switched on, the top electrode **504** and the bottom electrode **506** may be oppositely charged such that an electric field **512** is established therebetween. In some embodiments, the electric field **350** is a direct-current (DC) electric field. In alternative embodiments, the electric field **350** is an alternating-current (AC) electric field. Furthermore, the polarity of the electrodes **504** and **506** may be reversed for applications of the present disclosure. In some examples, the voltage supplied by the power source **508** is from about 2 kV to about 10 kV; though other voltage may also be applicable in the present disclosure. For embodiments in which the top electrode **504** and the bottom electrode **506** are substantially parallel to each other (FIG. **10C**), a direction of the electric field **350** may be substantially unidirectional, such as substantially perpendicular to both the top electrode **504** and the bottom electrode **506**. Alternatively, referring to FIG. **10D**, the direction of the electric field **350** may be multidirectional, i.e., the electric field **350** does not align along a single direction.

(50) In some embodiments, the mode by which the method **100** performs the annealing process is not limiting so long as it provides sufficient energy to enable the transformation of amorphous HfO.sub.2 to ferroelectric orthorhombic crystalline HfO.sub.2. For example, the annealing process may be a furnace annealing process, a rapid thermal annealing (RTA) process, a spike annealing process, a laser annealing process, other suitable annealing process, or combinations thereof. In an example embodiment, the annealing process is an RTA process. In some embodiments, the annealing temperature implemented during the annealing process ranges from about 600 degrees Celsius to about 1000 degrees Celsius. If the annealing temperature falls below about 600 degrees Celsius, the thermal energy available for completing the phase transformation may not be sufficient. On the other hand, if the annealing temperature is above about 1000 degrees Celsius, excessive heat may damage other components (e.g., S/D features, etc.) of the device **200**. The annealing process may be implemented for any suitable amount of time depending upon a specific annealing process employed. For example, if a high-temperature annealing process, such as a laser annealing process, is employed, the annealing time may be on the order of microseconds. On the other hand, if a low-temperature annealing process, such as a furnace annealing process, is employed, the annealing time may be on the order of tens of minutes.

(51) As discussed above, the deposition of the Hf-containing layer **232A** at operation **108A** may not necessarily be accompanied by the in-situ application of the electric field **320**. At operation **114**, however, the method **100** applies the electric field **350** while performing the annealing process. In some embodiments, parameters for implementing the electric field **350** are similar to those for implementing the electric field **320**. For example, the direction and strength (i.e., the magnitude) of the electric field **350** are substantially the same as those of the electric field **320**. In other embodiments, the electric field **350** may be implemented at a higher magnitude than the electric field **320**. As discussed above, applying the electric field **320** during deposition of HfO.sub.2 at operation **108A** provides energy for structural rearrangement of atoms within the structure of HfO.sub.2. One result of such rearrangement includes reduction in the number of oxygen vacancies present in HfO.sub.2. However, the simultaneous application of the electric field **350** during the annealing process at operation **114** generally serves a purpose different from that of the electric field **320** applied during the deposition process at operation **108A**. As stated previously, the activation energy required (i.e., energy barrier that must be overcome) for the crystallization of the ferroelectric orthorhombic phase increases as thickness of the high-k gate dielectric layer decreases. For at least this reason, thermal annealing alone, carried out at a temperature (from about 600 degrees Celsius to about 1000 degrees Celsius) high enough without causing damage to other

components of the device, falls short in converting amorphous HfO₂ to ferroelectric orthorhombic HfO₂. In the present disclosure, the simultaneous application of the electric field **350** during the annealing process is configured to overcome the activation energy required for crystallizing or transforming amorphous HfO₂ in the Hf-containing layers **234A** and **234B** to a ferroelectric orthorhombic phase (i.e., Pca2₁ phase) of HfO₂. In an example embodiment, the amount of ferroelectric orthorhombic phase obtained following the annealing process is at least about 60% by weight. In some embodiments, after the annealing process at operation **114**, the electric dipoles within the Hf-containing layers **234A** and **234B** are oriented in substantially the same direction. In further embodiments, a domain size of the ferroelectric Hf-containing material in the Hf-containing layers **234A** and **234B** is larger than the thickness *t*.

(52) FIG. **11** is a schematic illustration of the effect of applying an electric field (e.g., the electric field **350**) on the activation energy required for phase transformation during the annealing process. As activation energy is plotted against polarization in the depicted graph, the activation energy **610** at the peak of the graph reflects the amount of energy needed to overcome the barrier for forming the ferroelectric orthorhombic phase of HfO₂ without applying any electric field. On the other hand, the activation energy **620** represents the energy barrier against forming the ferroelectric orthorhombic phase of HfO₂ when an electric field, such as the electric field **350**, is applied during the annealing process. Notably, the activation energy **620** is less than the activation energy **610**, indicating a lowered energy barrier for forming the ferroelectric orthorhombic phase of HfO₂. In other words, although the chemical composition of the Hf-containing layer **234** is the same as that of the Hf-containing layers **232A** and **232B**, the Hf-containing layer **234** has a higher ferroelectricity in comparison to the Hf-containing layers **232A** and **232B**. In some examples, the activation energy **620** may be reduced to about 20% to about 25% of the activation energy **610**, thereby increasing the efficiency of HfO₂ transformation.

(53) Now referring to FIGS. **1**, **12**, and **13**, the method **100** at operation **116** forms a metal gate electrode **228** over the Hf-containing layer **234**, thereby completing the formation of the HKMG **210**. Though not depicted, in some embodiments, forming the metal gate electrode **228** includes forming various material layers such as a barrier layer, a capping layer, other suitable material layers, or combinations thereof over the Hf-containing layer **234**. In some examples, the barrier layer may include titanium nitride (TiN), tantalum nitride (TaN), other suitable materials, or combinations thereof, and the capping layer may include TiN, TaN, titanium silicide (TiSi), other suitable materials, or combinations thereof, both of which may be formed by any suitable method, such as CVD, ALD, PVD, other suitable methods, or combinations thereof.

(54) Thereafter, the method **100** deposits one or more work function metal layer over the Hf-containing layer **234**. The work function metal layer may include any suitable material, such as titanium nitride (TiN), tantalum nitride (TaN), ruthenium (Ru), molybdenum (Mo), tungsten (W), platinum (Pt), titanium (Ti), aluminum (Al), tantalum carbide (TaC), tantalum carbide nitride (TaCN), tantalum silicon nitride (TaSiN), titanium silicon nitride (TiSiN), other suitable materials, or combinations thereof. In some embodiments, the work function metal layer includes multiple material layers of the same (i.e., all n-type work function metals or all p-type work function metal) or different types in order to achieve a desired threshold voltage. For example, the work function metal layer may include an n-type work function metal layer formed over a p-type work function metal layer (or vice versa). The work function metal layer may be formed by any suitable method, such as CVD, ALD, PVD, other suitable methods, or combinations thereof.

(55) Subsequently, the method **100** forms a bulk conductive layer over the one or more work function metal layer. The bulk (or fill) conductive layer may include aluminum (Al), copper (Cu), tungsten (W), cobalt (Co), ruthenium (Ru), other suitable conductive materials, or combinations thereof, and may be formed by any suitable method, such as CVD, ALD, PVD, plating, other suitable methods, or combinations thereof. Thereafter, the method **100** may perform one or more polishing process (e.g., CMP) to remove any excess conductive materials and portions of the Hf-

containing layer **234B** formed over the ILD layer **218** to planarize the top surface of the device **200**. (56) The method **100** at operation **116** may also form other material layers. For example, the method **100** may form a hard mask layer before forming the work function metal layer to accommodate patterning of the work function metal layer. The hard mask layer may include any suitable material, such as a titanium-containing material, a tantalum-containing material, a nitrogen-containing material, a silicon-containing material, other suitable materials, or combinations thereof. The hard mask layer may, for example, include titanium nitride (TiN), tantalum nitride (TaN), titanium silicide (TiSi), and/or other suitable materials. The hard mask layer may be formed by any suitable deposition process (e.g., ALD). The method **100** may additionally form a glue (or adhesive) layer (not depicted) before forming the bulk conductive layer. In some embodiments, the glue layer is configured to prevent peeling of the bulk conductive layer (e.g., a W-containing bulk conductive layer). The glue layer may include a titanium-containing material, a tantalum-containing material, a tungsten-containing material, a nitrogen-containing material, other suitable materials, or combinations thereof. The glue layer may be formed by any suitable method (e.g., CVD, ALD, PVD, etc.).

(57) Referring to FIG. **1**, the method **100** at operation **118** may perform additional processing steps. For example, additional vertical interconnect features such as contacts and/or vias, and/or horizontal interconnect features such as lines, and multilayer interconnect features such as metal layers and interlayer dielectrics can be formed over the device **200**. The various interconnect features may implement various conductive materials including copper (Cu), tungsten (W), cobalt (Co), aluminum (Al), titanium (Ti), tantalum (Ta), platinum (Pt), molybdenum (Mo), silver (Ag), gold (Au), manganese (Mn), zirconium (Zr), ruthenium (Ru), their respective alloys, metal silicides, other suitable materials, or combinations thereof. The metal silicides may include nickel silicide, cobalt silicide, tungsten silicide, tantalum silicide, titanium silicide, platinum silicide, erbium silicide, palladium silicide, other suitable metal silicides, or combinations thereof.

(58) Now referring back to FIG. **1** and collectively to FIGS. **14A-14E**, the method **100** may alternatively proceed in Path B after forming the sacrificial layer **230**. Referring to FIG. **14A**, the method **100** at operation **108B** forms a Hf-containing layer **252A** over the fin **204** in the gate trench **220**. Similar to the process of forming the Hf-containing layer **232A** at operation **108A**, the Hf-containing layer **252A** is selectively formed over the fin **204** in a process facilitated by the sacrificial layer **230**. In some embodiments, the Hf-containing layer **252A** has a chemical composition substantially similar to that of the Hf-containing layer **232A**. For example, the Hf-containing layer **252A** may include a high-k dielectric material comprising hafnium (Hf) and oxygen (O) in the form of, for example, amorphous hafnium oxide (e.g., HfO₂) and may optionally be doped with one or more dopant such as, for example, Zr, Al, La, Ti, Ta, Si, Y, Sc, other suitable elements, or combinations thereof. The Hf-containing layer **252A** may be formed by any deposition process including ALD, CVD, other suitable processes, or combinations thereof.

(59) Notably, while the Hf-containing layer **252A** and the Hf-containing layer **232A** include substantially similar chemical compositions and are both substantially amorphous in nature, the microstructure of the Hf-containing layer **252A** may differ from that of the Hf-containing layer **232A**. This may be a result of the in-situ application of the electric field **320** during the forming of the Hf-containing layer **232A**. As discussed earlier, the in-situ application of the electric field **320** provides energy to mobilize oxygen atoms relative to hafnium atoms toward lattice positions consistent with the ferroelectric orthorhombic crystal structure, making the microstructure favorable for transforming amorphous HfO₂ into ferroelectric orthorhombic HfO₂ when thermal annealing is applied. In comparison, the HfO₂ in the Hf-containing layer **252A** lacks or substantially lacks the microstructure having an amount of oxygen vacancies favorable for phase transformation due to a lack of applied in-situ electric field. As such, in some examples, the Hf-containing layer **232A** includes less oxygen vacancies than the Hf-containing layer **252A**.

(60) Referring to FIGS. **1** and **14B**, the method **100** at operation **110** removes the sacrificial layer

230 from the device **200** in a process similar to that discussed above with reference to FIG. **9B**. For example, the method **100** at operation **110** may perform a wet etching process **330** utilizing HF to remove the sacrificial layer **230**. As discussed above, the removal of the sacrificial layer **230** forms trench **240** having a width t disposed between the Hf-containing layer **252A** and the gate spacers **212**.

(61) Thereafter, referring to FIGS. **1** and **14C**, the method **100** at operation **112B** deposits a first portion of the Hf-containing layer **252B** over the Hf-containing layer **252A** and a second portion of the Hf-containing layer **252B** on sidewalls of the gate spacers **212**. In the depicted embodiment, the Hf-containing layer **252A** and the first portion of the Hf-containing layer **252B** are together referred to as Hf-containing layer **253**. As depicted herein, the thickness t of the Hf-containing layer **252B** is substantially equivalent to the width t of the trench **240**. The deposition process at operation **112B** may be similar to that discussed above with respect to operation **112A**; however, unlike at operation **112A**, no in-situ electric field is applied at operation **112B**. The Hf-containing material may be deposited by any deposition process such as ALD, CVD, other suitable processes, or combinations thereof. As such, microstructure of the Hf-containing material in the Hf-containing layers **252A** and **252B** differs from that in the Hf-containing layers **232A** and **232B** for reasons similar to those discussed above with respect to operation **108B**.

(62) Referring to FIGS. **1** and **14D**, the method **100** at operation **114** performs an annealing treatment while applying an electric field **350** to the device **200**. In some embodiments, the method **100** at operation **114** transforms amorphous $\text{HfO}_{2.2}$ included in the Hf-containing layer **253** and the Hf-containing layer **252B** into ferroelectric orthorhombic $\text{HfO}_{2.2}$ to form the Hf-containing layers **234A** and **234B**, respectively. In some embodiments, the method **100** implements operation **114** in a manner substantially similar to that discussed above with reference to FIG. **9D** (i.e., at Path A).

(63) Notably, the thickness T of the Hf-containing layer **234A** is greater than the thickness t of the Hf-containing layer **234B**. In the depicted embodiment, the thickness T is about 1 nm to about 3 nm similar to that discussed above. In some examples, a ratio of T to t may be at least 3. Of course, the present disclosure is not limited to such ratio. In some instances, it may be desirable to minimize the thickness t (i.e., maximizing the ratio of T to t) to an extent allowable by the capacity of the processing tool(s) as discussed above. As such, because the thickness of the Hf-containing layer **234B** is less than the thickness of the Hf-containing layer **234A**, parasitic capacitance on the spacers **212** is effectively reduced. Furthermore, reduced thickness of the Hf-containing layer **234B** enlarges a width w of the gate trench **220**, thus enabling the formation of a larger gate electrode (e.g., subsequently formed gate electrode **228**) with reduced resistance and/or improved tuning of threshold voltage.

(64) Referring to FIGS. **1** and **14E**, the method **100** at operation **116** forms the metal gate electrode **228** over the Hf-containing layers **234A** and **234B**, completing the formation of the HKMG **210** in a manner substantially similar to the discussion above with respect to FIG. **12**. For example, forming the metal gate electrode **228** may include forming various material layers such as a barrier layer, a capping layer, at least one work function metal layer, a bulk conductive layer, other suitable material layers, or combinations thereof over the Hf-containing layers **234A** and **234B**. The various material layers of the metal gate electrode **228** may be formed by any suitable method including CVD, ALD, PVD, plating, other suitable methods, or combinations thereof. A CMP process may be performed after forming the metal gate electrode **228** to planarize the top surface of the device **200** (e.g., removing portions of the Hf-containing layer **234B** formed over the ILD layer **218**).

Thereafter, the method **100** at operation **118** performs additional processing steps such as forming a variety of interconnect features over the device **200** as discussed in detail above.

(65) Although not intended to be limiting, one or more embodiments of the present disclosure provide many benefits to a semiconductor device and the formation thereof. The present disclosure provides methods of forming an HKMG in an FET such as an NCFET. In some embodiments, the

HKMG includes a ferroelectric Hf-containing (e.g., HfO₂-containing) high-k gate dielectric layer fabricated by a series of deposition and annealing processes. Particularly, methods of transforming amorphous HfO₂-containing dielectric material to ferroelectric orthorhombic HfO₂-containing dielectric layer (i.e., including the Pca21 phase) include simultaneously applying an electric field during a deposition and/or a post-deposition annealing process. In addition, embodiments of the present disclosure are also directed to methods of reducing sidewall thickness of the ferroelectric Hf-containing high-k gate dielectric layer in NCFETs. Embodiments provided herein improve the overall performance of devices such as NCFETs by promoting formation of ferroelectric Hf-containing gate dielectric layer, reducing parasitic capacitance between the HKMG and gate spacers, enlarging processing window for tuning threshold voltage, and reducing overall gate resistance of the devices.

(66) In one aspect, the present disclosure provides a method that begins with forming a sacrificial layer on sidewalls of gate spacers disposed over a semiconductor layer. The method then proceeds to forming a first hafnium-containing gate dielectric layer over the semiconductor layer in a first trench disposed between the gate spacers, removing the sacrificial layer to form a second trench between the gate spacers and the first hafnium-containing gate dielectric layer, and subsequently forming a second hafnium-containing gate dielectric layer over the first hafnium-containing gate dielectric layer and on the sidewalls of the gate spacers. Then, the method proceeds to annealing the first and the second hafnium-containing gate dielectric layers while simultaneously applying an electric field, and subsequently forming a gate electrode over the annealed first and second hafnium-containing gate dielectric layers.

(67) In another aspect, the present disclosure provides a method that begins with removing a dummy gate stack disposed over a semiconductor layer and interposed between gate spacers to form a first trench and forming a sacrificial layer on sidewall surfaces of the gate spacers. After forming the sacrificial layer, the method proceeds to depositing a hafnium-containing dielectric material over the semiconductor layer to form a first gate dielectric layer, where the depositing includes applying a first electric field. After forming the first gate dielectric layer, the method proceeds to removing the sacrificial layer to form a second trench and subsequently depositing the hafnium-containing dielectric material in the second trench and on the sidewall surfaces of the gate spacers to form a second gate dielectric layer, where the depositing includes applying a second electric field. Thereafter, the method proceeds to applying an annealing treatment to the first and the second gate dielectric layers, where the applying increases degree of ferroelectricity of the first and the second gate dielectric layers. After applying the annealing treatment, the method proceeds to forming a metal gate electrode over the first and the second gate dielectric layers.

(68) In yet another aspect, the present disclosure provides a semiconductor structure that includes gate spacers disposed over a semiconductor layer, a hafnium-containing dielectric layer, and a metal gate electrode disposed over the hafnium-containing dielectric layer and between the gate spacers. With respect to the hafnium-containing dielectric layer, the present disclosure provides that a first portion of the hafnium-containing dielectric layer having a first thickness is disposed over the semiconductor layer and a second portion of the hafnium-containing dielectric layer having a second thickness is disposed on sidewalls of the gate spacers, where the first thickness is greater than the second thickness.

(69) The foregoing has outlined features of several embodiments so that those skilled in the art may better understand the detailed description that follows. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure, comprising: a semiconductor layer disposed over a substrate; a shallow trench isolation (STI) structure disposed over the substrate and surrounding the semiconductor layer; gate spacers disposed over the semiconductor layer; an interfacial layer disposed on the semiconductor layer; a hafnium-containing dielectric layer of a same and uniform material in its entirety, wherein a first portion of the hafnium-containing dielectric layer having a first thickness is disposed on the interfacial layer and a second portion of the hafnium-containing dielectric layer having a second thickness is disposed along sidewalls of the gate spacers and interfacing side surfaces of a metal gate electrode, and wherein the first thickness is greater than the second thickness; the metal gate electrode disposed over the hafnium-containing dielectric layer and between the gate spacers; and source/drain features formed over the semiconductor layer, the source/drain features directly contact side surfaces of the gate spacers, wherein a top surface of the STI structure is below a bottom surface of the source/drain features, wherein bottom surfaces of the interfacial layer, the second portion of the hafnium-containing dielectric layer, and the gate spacers are substantially coplanar, wherein the hafnium-containing dielectric layer includes a ferroelectric orthorhombic phase in both the first and second portions of the hafnium-containing dielectric layer, wherein electric dipoles within both the first and second portions of the hafnium-containing dielectric layer are oriented substantially in a same direction parallel to an electric field that is applied when forming the first and second portions of the hafnium-containing dielectric layer.
2. The semiconductor structure of claim 1, wherein the hafnium-containing dielectric layer includes oxygen vacancies.
3. The semiconductor structure of claim 1, wherein the second portion of the hafnium-containing dielectric layer separates the interfacial layer from the gate spacers.
4. The semiconductor structure of claim 1, wherein a ratio of the first thickness to the second thickness is at least about 3.
5. The semiconductor structure of claim 1, wherein the second portion of the hafnium-containing dielectric layer is spaced apart from the source/drain features by the gate spacers such that a distance between the second portion of the hafnium-containing dielectric layer and the source/drain features is substantially equal to a width of the gate spacers.
6. The semiconductor structure of claim 1, wherein the applied electric field is unidirectional with a direction substantially perpendicular to a top surface of the substrate.
7. The semiconductor structure of claim 1, wherein the applied electric field is multidirectional at directions not aligned along any single direction.
8. The semiconductor structure of claim 1, wherein a top surface of the source/drain features is below a top surface of the first portion of the hafnium-containing dielectric layer.
9. A semiconductor structure, comprising: a semiconductor fin disposed over a substrate; an isolation structure disposed over the substrate and surrounding the semiconductor fin; a metal gate structure disposed over the semiconductor fin, wherein the metal gate structure includes an interfacial layer, a hafnium oxide layer disposed over the interfacial layer, and a metal gate electrode disposed over the hafnium oxide layer, wherein the hafnium oxide layer is of a same and uniform material in its entirety; a gate spacer disposed along a sidewall of the metal gate structure, wherein a first portion of the hafnium oxide layer disposed along the gate spacer and interfacing side surfaces of the metal gate structure is thinner than a second portion of the hafnium oxide layer disposed directly over the interfacial layer, and a bottom surface of the first portion of the hafnium oxide layer is below a bottom surface of the second portion of the hafnium oxide layer; a source/drain feature disposed over the semiconductor fin and adjacent the gate spacer; and an interlayer dielectric (ILD) layer disposed over the source/drain feature and the isolation structure, wherein a bottom surface of the ILD layer is below a bottom surface of the source/drain feature,

wherein electric dipoles within both the first and second portions of the hafnium oxide layer are oriented substantially in a same direction unidirectional and perpendicular to a top surface of the substrate.

10. The semiconductor structure of claim 9, wherein the interfacial layer is disposed directly between the semiconductor fin and the second portion of the hafnium oxide layer.

11. The semiconductor structure of claim 10, wherein the first portion of the hafnium oxide layer extends to contact sidewalls of the interfacial layer.

12. The semiconductor structure of claim 9, wherein the hafnium oxide layer includes hafnium oxide in a ferroelectric orthorhombic phase.

13. The semiconductor structure of claim 9, wherein the first portion of the hafnium oxide layer has a first thickness and the second portion of the hafnium oxide layer has a second thickness, and wherein a ratio of the first thickness to the second thickness is no greater than 1:3.

14. The semiconductor structure of claim 9, wherein the metal gate electrode includes a bulk conductive layer disposed over a work function metal layer.

15. The semiconductor structure of claim 9, wherein the second portion of the hafnium oxide layer is directly on the interfacial layer, and wherein the metal gate electrode is in direct contact with both the first portion of the hafnium oxide layer and the second portion of the hafnium oxide layer.

16. The semiconductor structure of claim 13, wherein the second thickness of the second portion of the hafnium oxide layer is between about 1 nm and about 3 nm.

17. A semiconductor structure, comprising: a semiconductor fin disposed over a substrate; a shallow trench isolation (STI) layer disposed over the substrate and surrounding the semiconductor fin; a metal gate structure disposed over the semiconductor fin, wherein the metal gate structure includes: an interfacial layer; a hafnium oxide layer disposed over the interfacial layer, wherein the hafnium oxide layer is of a same and uniform material in its entirety; and a metal gate electrode disposed over hafnium oxide layer; and a gate spacer disposed along a sidewall of the metal gate structure, wherein a first portion of the hafnium oxide layer disposed along the gate spacer and interfacing side surfaces of the metal gate structure has a first thickness and a second portion of the hafnium oxide layer disposed along the interfacial layer has a second thickness that is greater than the first thickness; a source/drain epitaxial feature disposed over the semiconductor fin and adjacent the metal gate structure; and an interlayer dielectric (ILD) layer disposed over the source/drain epitaxial feature and the STI layer, wherein the source/drain epitaxial feature is isolated from the STI layer by the ILD layer, wherein a top surface of the semiconductor fin is in direct contact with the interfacial layer, the second portion of the hafnium oxide layer, and the gate spacer, wherein both the first and second portions of the hafnium oxide layer are ferroelectric having a crystalline structure, wherein electric dipoles within both the first and second portions of the hafnium oxide layer are oriented substantially in a same direction parallel to each of a first applied electric field when forming the second portion of the hafnium oxide layer, a second applied electric field when forming the first portion of the hafnium oxide layer, and a third applied electric field when annealing the first and second portions of the hafnium oxide layer.

18. The semiconductor structure of claim 17, wherein the interfacial layer is separated from the gate spacer by the first portion of the hafnium oxide layer.

19. The semiconductor structure of claim 17, wherein a ratio of the second thickness to the first thickness is at least about 3.

20. The semiconductor structure of claim 17, wherein both the first and second portions of the hafnium oxide layer includes hafnium oxide in an orthorhombic phase.
